

An Exploratory Analysis of 2014-15 NBA data

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Abstract

This report presents exploratory statistical analyses of data from the 2014-15 NBA season of the National Basketball Association (NBA). We aim to provide an effective overview of factors that are not purely about the players, but also about the circumstances under which they play. In addition to player skills, the report examines how variables such as location, scheduling, and game context can influence outcomes by investigating factors such as home advantage, rest days, and clutch situations. The report highlights the importance of environmental and psychological factors alongside physical talent and technical ability. It seeks to explore the simple question, ‘What drives success in professional basketball?’

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1 Introduction and data preparation

This report presents our statistical analyses performed using data from the 2014-15 season of the National Basketball Association (NBA). We consider data from `Shot_data_Clean.csv` and `Players_clean.csv`, which are cleaned versions of data available in the [NBA-shot-logs entry on the Kaggle website](#).

Before conducting the analysis, we cleaned and prepared the data from Kaggle so that the variables used in the report are more consistent and reliable. After these cleaning operations, we obtain `Shot_data_clean.csv` containing 124,343 shot attempts and 23 features, covering 903 games.

We also use `Players_clean.csv`, which gives player information such as team, position and experience. We had to remove the duplicates as well as change some elements in experience from R to 0, so we can use them later in our analyses.

This report is organised into 4 main sections. Section 2 contains our investigation into clutch shots as well as changes in performance at different point differences. In Section 3, we explore whether home advantage exists and the possible causes relating to accuracy, experience, touch time, and number of shots attempted. Section 4 studies if there is a correlation between the number of days rest, and a team's win percentage. In Section 5, we analyse the impact of stars and role players both offensively and defensively. Section 6 includes a discussion of our overall findings.

2 What is a clutch shot?

In sports, it is very typical that pressure causes players to perform worse. While our preliminary data showed that the overall shot success rate drops during clutch time, we wanted to investigate why this happens and whether the actual value of the shots taken changes.

2.1 Performance vs Value

We first evaluated whether a player's overall shooting accuracy declines under late-game pressure. Using our filtered pool of reliable players (those with 5 or more clutch attempts), we aggregated the total makes and attempts for both clutch and normal play. A two-proportion hypothesis test was conducted to compare these overall success rates.

The test yielded a p-value of ≈ 0.0126 . This p-value being smaller than the standard 0.05 (and even stricter 0.025) threshold indicates that the difference is statistically significant. During normal play, these players convert shots at a rate of roughly 45.3%. However, during clutch time, this success rate drops significantly to 42.9%. The data suggests that players play significantly worse, experiencing a measurable drop in overall accuracy when the game is on the line.

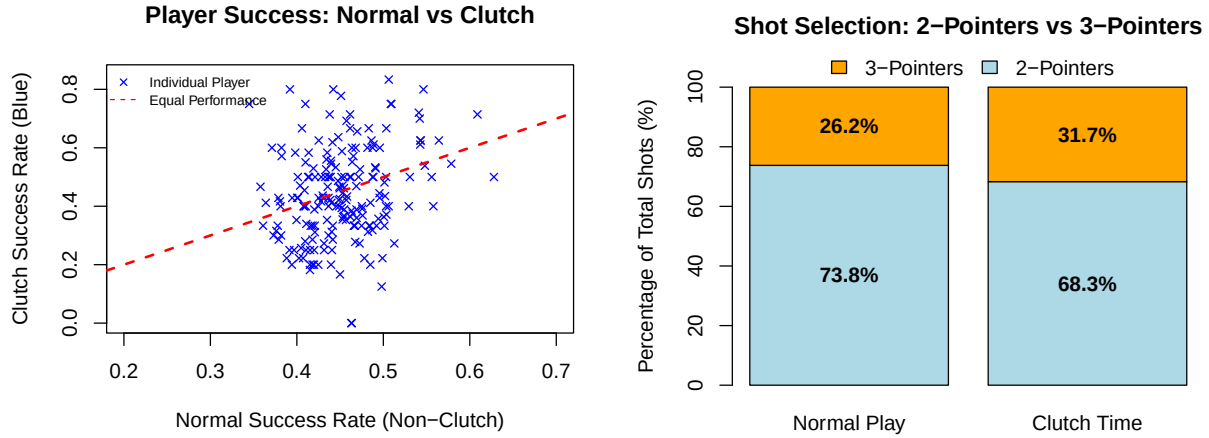
This universal "clutch penalty" is visualised in Figure 1a. When plotting individual player success rates, the majority of our reliable players fall below the red equality line, confirming a personal drop in shooting accuracy during high-pressure moments.

Despite this clear drop in shooting accuracy, we must also consider the actual value of the shots being taken. To investigate whether this drop in accuracy harms the team's expected point yield, we calculated the points yielded per shot (`Clutch_shot_ratio`) for each player and performed a paired t-test comparing their clutch ratio to their normal game ratio.

```
## [1] 0.4425747
```

As shown by the p-value of ≈ 0.44 generated from the paired t-test, the difference in points yielded per shot is not statistically significant.

This presents an intriguing paradox: despite a proven drop in overall shooting accuracy during the clutch, the actual value of those shots remains stable. This stability suggests a strategic shift in shot selection, pointing to the hypothesis that players compensate for the pressure by attempting higher-value, lower-percentage 3-point shots at a much greater rate.



(a) Individual player success rates: Normal Play vs Clutch Time. (b) Comparison of shot selection (2-pointers vs 3-pointers).

Figure 1: Analysis of player performance under pressure.

2.2 The Shift in Shot Selection

```
prop.test(x = c(clutch_3pt_attempts, normal_3pt_attempts),
          n = c(total_clutch_shots, total_normal_shots))$p.value

## [1] 3.607864e-11
```

To explain this paradox, we hypothesised that players might be changing their offensive strategy under pressure, specifically by settling for higher-value but lower-percentage 3-point shots. We implemented a two-proportion test to compare the rate of 3-point attempts in normal situations versus clutch situations.

The results of this shift in shot selection are presented in Figure 1b. The statistical test yielded a p-value of < 0.0001 , indicating that the difference is statistically significant. The data suggests that players intentionally shift their shot selection during clutch time, attempting 3-pointers at a higher rate (roughly 31.7%) compared to normal play (26.2%). Because 3-pointers are inherently more difficult, this strategic shift naturally lowers their overall shooting percentage, while the extra point awarded for a successful make mathematically offsets the difference, keeping the expected Points Per Shot stable.

2.3 Exploring the Components of a Clutch Shot

After investigating clutch performance, we were then motivated to explore the effect of score margins on shooting performance more widely, rather than just in the final 5 minutes. During this investigation, we will use 2 measures of shooting performance: Shot success average: A measure of how many of their shots they make on average points per shot ratio (PPS): A measure of how many points they score per shot on average

2.3.1 Testing

Initially, we planned to categorise margins into two situations: close ($|m| \leq 5$) and large ($|m| > 5$). However, we realised that a more realistic approach would be to introduce a third category by splitting big-margin situations based on whether the shooter's team was winning or losing, as this factor would affect a shot's context.

Our null hypothesis was that margin has an effect on shooting performance. Based on our earlier clutch shot analysis, we expected that players would perform better when the margin is large compared to when it is close. However, the specific impact of being on the winning versus losing team in large-margin situations was uncertain.

To begin testing this hypothesis, we calculated each player's shot success average (SSA) and points per shot (PPS) for each margin category. To assess whether there were statistically significant differences between the three conditions, we conducted a Friedman test for both SSA and PPS, as these involve repeated measures across multiple related samples.

SSA p-value

```
## [1] 4.487804e-63
```

PPS p-value

```
## [1] 3.428299e-68
```

For both measures, the p-values were less than 0.05, indicating a statistically significant difference in performance across margin conditions. Having established the presence of an overall difference, we used pairwise Wilcoxon signed-rank tests to identify which specific conditions differed, whilst comparing mean values to determine the direction of these differences.

2.3.2 Results and Interpretation

The differences between each margin condition were found to be statistically significant, with the results revealing several interesting patterns, visualised below using box plots.

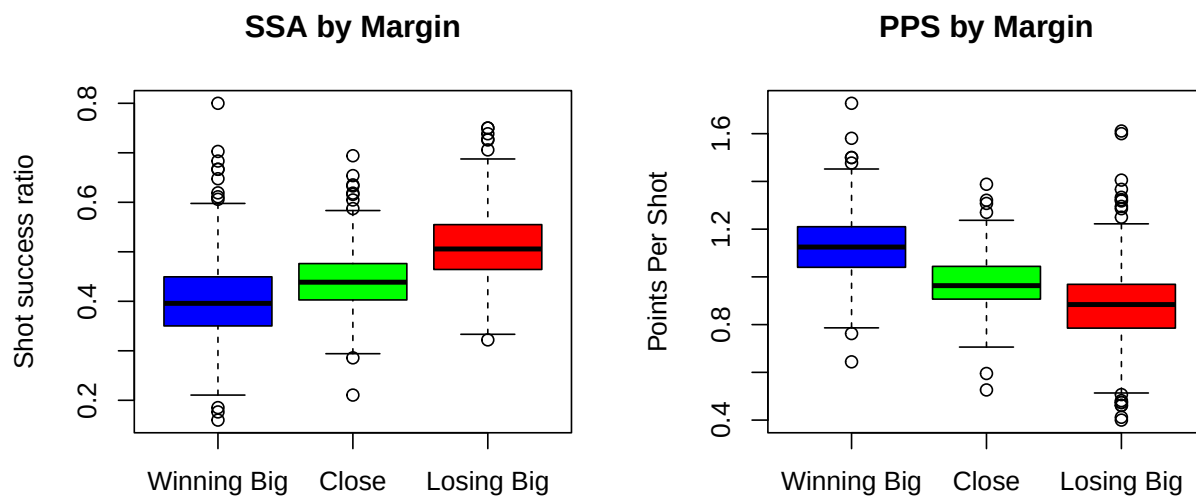


Figure 2: (Left) Boxplot of players' SSA by margin situation. Shot accuracy is highest when the team is losing with a large margin, followed by a close margin and then winning with a big margin. (Right) Boxplot of players' PPS ratio by margin situation. Points per shot are highest when the team is winning by a large margin, followed by a close margin and then losing with a big margin.

Figure 2 (left) suggests that, on average, players were less accurate when their team was leading by a large margin. Several factors may explain this outcome, including both psychological and tactical influences. For example, players may become more relaxed when their team is comfortably ahead, leading to reduced concentration, precision, and timing. Additionally, players may attempt riskier shots from larger distances. Teams in a strong leading position may also focus more on game management, such as running down the clock or avoiding injuries, rather than maximising shooting accuracy.

We then repeated the pairwise analysis using points per shot (PPS) for each player. Again, the differences between margin conditions were statistically significant.

Notably, the average ordering seen in Figure 2 (right) is the reverse of that observed for SSA, which was an unexpected result. This suggests that although players on the losing team shoot with higher accuracy, they may be taking lower-value shots on average. Due to less distance, two-point shots are generally easier and have a higher success rate than three-point shots. Therefore, it is plausible that players on a losing team prioritise more efficient yet lower-value scoring opportunities, while players on a winning team take on higher-risk, higher-reward shots.

While large margin shooting performance showed variation in our exploration, average close margin shooting sits in the middle of both SSA and PPS ranking, perhaps suggesting it's a balanced game state where opposing effects cancel out. For instance, in a close margin, teams have both high pressure (which can decrease performance) and high focus (increasing performance), cancelling out. There may also be a trade-off between risk and reward, as teams cannot take overly risky shots (as they did when comfortably winning) but also want higher-value shots (in order to pull ahead), perhaps explaining the moderate accuracy (SSA) and efficiency (PPS) seen in this situation.

3 Home advantage and causes in the NBA

In Subsection 3.1, we investigate whether a home advantage exists and the significance of the difference between home and away win rates. Then, in Subsection 3.2, we look at shot accuracy home and away, and how the experience of players affects this, as well as the difference in the number of team wins at home and wins away against the average experience of each team. Finally, in Subsection 3.3, we will explore whether teams at home have shorter touch times and take more shots, which may result in more points scored.

3.1 Existence of home advantage

We use data such as `GAME_ID`, `LOCATION`, and `WIN_LOSE` to determine whether the home team won or lost for each unique game. Then, we can test if the ratio of wins to total games is greater than the expected probability of 0.5. We do this via a binomial test with $H_0 : p_H = 0.5$, $H_1 : p_H > 0.5$

```
## home_results
##      L      W
## 397 506

## The binomial test p-value is 0.000160
## The estimate of the mean is 0.560354
```

The p-value we obtain (≈ 0.0002) is exceptionally small and at a 5% significance level, we reject H_0 in favour of H_1 . Our calculated home win rate is approximately 56% which is a large difference compared to our null hypothesis 50%. To conclude, we observe there is an increase in win rate at home and in the upcoming subsections we will explore possible reasons for the increase.

3.2 Shot success rates and impact of experience on home advantage

Firstly, we must consider if players are more effective at scoring both two and three point shots when at home to away, as this directly contributes to a team's effectiveness. To test this we will create two ratios for each player, calculated by dividing that player's home accuracy by the away accuracy for two pointers and three pointers. We will perform a t-test, with the alternative hypothesis being that players perform better at home (the ratio for players being > 1)

```
## p-value, mean estimate from t-test of two point ratios:
##          0.004175, 1.027744

## p-value, mean estimate from t-test of three point ratios:
##          0.003888, 1.117413
```

The very low p-values indicate that it is highly likely that players have a higher shot accuracy at home than away, suggesting that one contributing factor to the higher win rate at home is the higher shot accuracy at home, positively influencing the amount of points scored in a game. Additionally, the mean of the three point ratios is greater than the mean of the two point ratios, suggesting that the shot accuracy of three pointers is increased more by playing at home than that of two pointers.

We now aim to check if players shot accuracy increases as their experience increases. If there is a positive correlation, this could suggest that more experienced players have spent more time at home and, therefore, increased the ratio of shots made at home. This is achieved by using information such as `EXPERIENCE`, `LOCATION`, `PTS_TYPE`, and `SUCCESS`. For each player, we calculate the ratio of home accuracy to away accuracy for two pointers and three pointers and plot these against experience. We also compute a Pearson's product-moment correlation test for each with $H_0 : \rho_H = 0$, $H_1 : \rho_H \neq 0$.

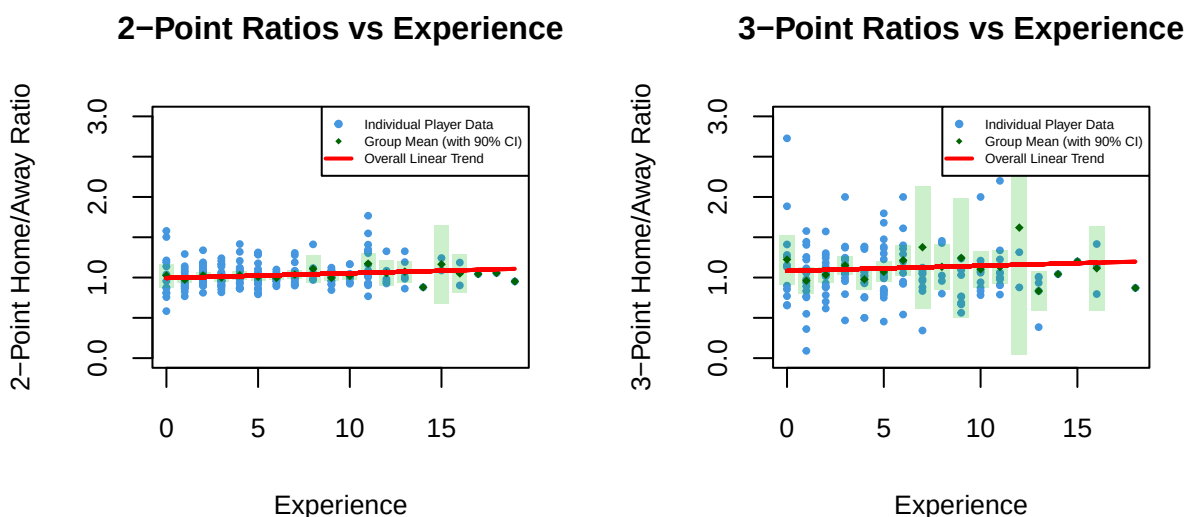


Figure 3: Relation between player experience and shot accuracy ratio of home to away. The ratio is plotted for each player, with 0 representing the rookies. 90% confidence intervals for each group mean are also displayed. The data is split into the two (left) and three (right) point accuracy ratios.

```
## p-value, mean estimate from PMCC of two point ratios:
##          0.161754, 0.016089
```

```
## p-value, mean estimate from PMCC of three point ratios:
##          0.045027, 0.555210
```

For two pointers, our low p-value and low ρ suggests a weak positive correlation as is shown in figure 1 (left). However, for three pointers, we see that our p-value is very large, suggesting we should accept H_0 , and assume there is no correlation between players' experience and impact of home advantage on three point accuracy.

Next, we explore the impact of a team's average experience on if a team performs better at home, to test whether increased average experience contributes to home advantage. We did this by comparing each teams wins minus their wins away to their average experience. We again used a correlation test to explore the presence of a link between the average experience and the wins difference. $H_0 : \rho_H = 0, H_1 : \rho_H \neq 0$.

```
cor.test(team_exp_and_home_away_wins_diff$home_away_wins_diff,
         team_exp_and_home_away_wins_diff$Average_experience)$p.value
## [1] 0.6252058
```

Thus, at a 5% significance level we again accept H_0 , suggesting that the data we have is not statistically significant and they are uncorrelated.

We see that individuals, on average, have greater accuracy at home compared to away, which shows us that home advantage comes from player success. From what we explored, we can see no significant correlation between player experience and home advantage as both tests, accuracy and win difference, show no noticeable trend which might suggest a reason for the better performance. This must mean the cause of home advantage must lie elsewhere.

3.3 Impact of touch time and shots attempted on home advantage

By considering the shots from `Shots_clean` and then separating shots that were taken at home from those taken away and sorting the shots in each category by player, we can compute the number of shots taken at home and compare to the number of shots taken away for each player. After implementing a Kolmogorov-Smirnov test, we can determine whether there is a significant difference in the distribution of the number of shots taken for home and away players, providing evidence of whether being at home has an impact on the number of shots taken by the players.

```
ks.test(player_shot_count_home, player_shot_count_away)$p.value
## [1] 0.9173611
```

The obtained p-value is very high (> 0.9), indicating that there is no significant difference in the distribution of the number of shots taken for home and away players. This suggests that being at home does not have a significant impact on the number of shots taken by the players, so the existence of home advantage as demonstrated in Subsection 3.1 is unlikely to be caused by players attempting more shots when playing at home.

As the evidence suggests there is no difference in number of shots taken, we will next consider if more two or three pointer shots specifically are attempted when home, which, if true, may result in teams scoring higher-point shots, explaining the difference in home win rates and away win rates even if the number of shots attempted is very similar. To test this, we apply another Kolmogorov-Smirnov test, this time only considering shots where `PTS_TYPE` is either two or three. $H_0 : \mu_H = \mu_A, H_1 : \mu_H > \mu_A$

```
ks.test(player_shot_count_home_2pt, player_shot_count_away_2pt,
        alternative = 'greater')$p.value
```

```
## [1] 0.694222
```

```
ks.test(player_shot_count_home_3pt, player_shot_count_away_3pt,
        alternative = 'greater')$p.value
```

```
## [1] 0.643003
```

Much like the first test on shot attempt counts, the high p-values suggest accepting the null hypothesis, which states there is no difference in distribution for two or three point shot attempts home and attempts away.

Further investigating the factors that may contribute to the presence of home advantage, we can analyse TOUCH_TIME to consider the time players spent holding the ball before making a shot. Implementing another Kolmogorov-Smirnov to test, we aim to deduce if the distributions of the touch times differ, specifically if touch time is lower at home.

```
ks.test(home_touch_times, away_touch_times, alternative = 'greater')$p.value
```

```
## [1] 0.005676768
```

The very low p-value indicates there is a significant likelihood that the touch time of shots made away are greater than the touch times at home, which would indicate that players hold the ball for a shorter period of time before shooting when at home. It can be speculated that this is due to increased confidence and comfort when playing at home, or possibly due to the more convenient conditions of playing at home, such as not needing to travel long distances before the games.

4 Team Rest vs Team Win Percentage

We decided to investigate whether the number of days rest had any effect on team win-rates. To evaluate this, we categorised the variable Rest_days into four groups: 1 day, 2 days, 3 days, and 4+ days.

4.1 Statistical Summary of Win Rates

Summary statistics show that win rates across all groups cluster near the league average of 50%. We calculated the 95% confidence interval where $\hat{p}$ is the sample win rate and n is the number of games. The full confidence interval is then given by:

$$CI = \hat{p} \pm 1.96 \cdot \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}.$$

Rest_group	N	Win_per	CI_lwr	CI_upp
1 day	427	52.0	47.3	56.7
2 days	987	49.8	46.7	53.0
3 days	254	47.2	41.1	53.4
4+ days	108	50.0	40.6	59.4

Every confidence interval overlaps the 50% win-rate threshold, indicating that the minor fluctuations observed between groups are statistically indistinguishable from random variation. For example, the slight dip following 1 day of rest and the lower win rate in the 4+ day group do not provide clear evidence of a meaningful rest effect.

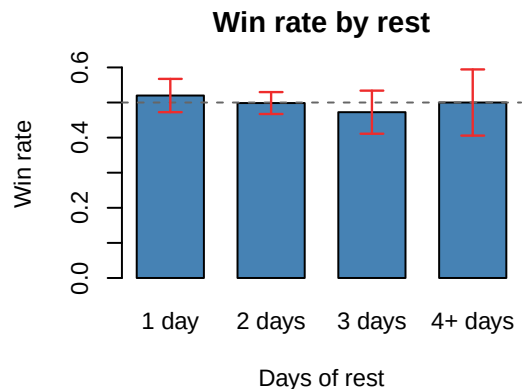


Figure 4 : Win percentage by number of days rest.

4.2 Statistical Inference: Pearson's Chi-Squared & Monte Carlo Simulation Test

To formally test for independence between rest and win probability, we performed a Pearson's chi-squared test on the contingency table of results. The hypotheses were:

- H_0 : Team win rate is independent of the number of rest days.
- H_1 : There is a significant association between rest days and the probability of winning.

The Pearson's chi-squared test statistic is calculated as:

$$\chi^2 = \sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i}$$

where O_i represents the observed frequencies in the contingency table and E_i represents the expected frequencies under H_0 .

Pearson's Chi-squared test

```
data: Cont_table
X-squared = 1.4576, df = 3, p-value = 0.6921
```

The test yielded $\chi^2 = 1.4576$ with $df = 3$ and a p-value of 0.6921. Since $p > 0.05$, we fail to reject the null hypothesis.

A Monte Carlo permutation test further supported this conclusion with a p-value of 0.4022. These results indicate that rest intervals do not significantly alter single-game win probabilities in this data set.

```
## observed_range_of_group_win_rates monte_carlo_p_value
##                                0.0475                0.6756
```

5 Two-Way Player Impact

Two-way players are defined by their ability to provide value on both offence and defence. This part of the analysis identifies high impact players and compares different player tiers.

5.1 Methodology and Metrics

Offensive efficiency was measured using points per shot (PPS), which incorporates the added value of three-point field goals. Defensive impact was estimated using opponent field goal percentage during a player's estimated on - window.

The impact score was defined as:

$$\text{Impact Score} = \text{PPS} - \text{Opp. FG\%}.$$

A higher score represents a player who scores efficiently while also suppressing opponent shooting. The top-ranked players in this analysis were Deandre Jordan (0.97), Tyson Chandler (0.913), and Kyle Korver (0.891).

5.2 Stars and Role Players

Players were classified by shot volume. Those in the 75th percentile or higher were labelled as stars, while the remaining players were labelled as role players. Below are the stars and role player tables respectively:

Table 1: Top 10 Stars by Impact Score

Player	Shots	PPS	Opp miss	Impact
Stephen Curry	945	1.182	0.568	0.750
Wesley Matthews	727	1.136	0.559	0.695
Klay Thompson	952	1.117	0.562	0.679
Al Horford	679	1.108	0.560	0.667
JJ Redick	625	1.118	0.546	0.664
Goran Dragic	746	1.102	0.542	0.643
Derrick Favors	674	1.089	0.548	0.637
Anthony Davis	829	1.094	0.535	0.629
Serge Ibaka	726	1.048	0.565	0.613
Chris Paul	877	1.068	0.544	0.612

Table 2: Top 10 Role Players by Impact Score

Player	Shots	PPS	Opp miss	Impact
Deandre Jordan	389	1.429	0.541	0.970
Tyson Chandler	334	1.359	0.553	0.913
Kyle Korver	444	1.327	0.565	0.891
Greg Smith	47	1.234	0.631	0.865
Luke Babbitt	153	1.275	0.533	0.808
Rudy Gobert	266	1.248	0.556	0.804
James Johnson	306	1.265	0.528	0.793
Ed Davis	301	1.223	0.525	0.748
Chris Andersen	135	1.193	0.551	0.744
Jared Dudley	344	1.174	0.565	0.739

Welch Two Sample t-test

data: impact_score by player_type

t = 0.90372, df = 190.43, p-value = 0.3673

A Welch two sample t-test was used to determine whether a significant gap exists between these groups. The test produced a p-value of 0.367, indicating no statistically significant difference in two-way impact between stars and role players. While stars face heavier offensive burdens, role players often benefit from higher efficiency, assisted shot opportunities.

6 Conclusion

Through our exploratory analysis of the 2014-15 NBA season, we uncovered several conclusions:

- While overall shooting accuracy drops in the clutch, the expected value (points per shot) remains stable. We demonstrated that this is driven by a statistically significant shift in shot selection, as players attempt harder 3-point shots at a higher rate when the game is on the line.
- After considering many possible causes for home advantage, shot accuracy, especially three point shot accuracy, is notably improved when at home. Touch time is also reduced at home, suggesting most of the impact is due to increased comfort, and confidence when at home.
- In light of this analysis, we can deduce that the number of days rest a team has does not significantly dictate independent game outcomes. This lack of statistical correlation emphasises the remarkable efficiency of the 82-game NBA schedule.
- While two-way efficiency helps us identify standout performers, there isn't a meaningful difference between stars and role players.

Our observations are supported by evidence and statistical testing, however, our conclusions could be strengthened by considering additional data such as unassisted shots, free throw, travel distance, and substitution data.